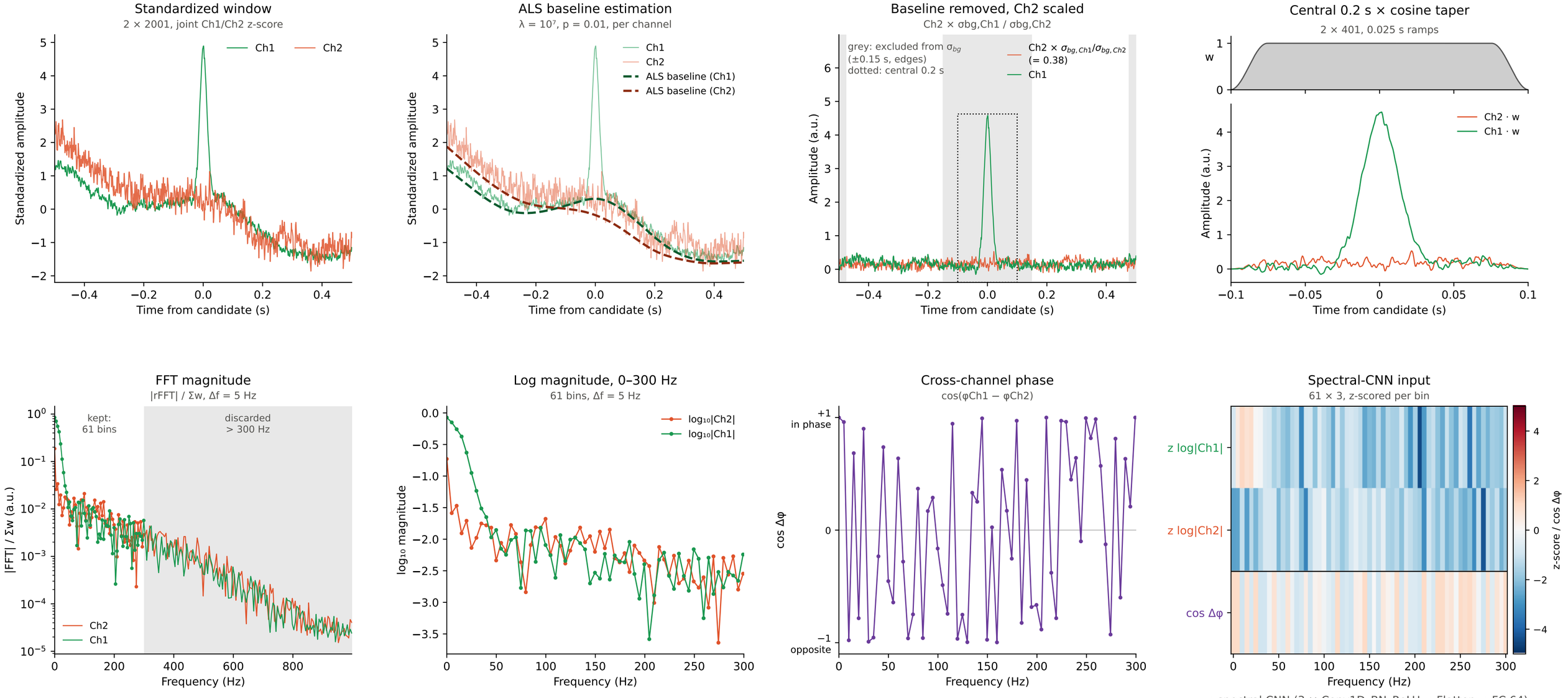

Standardized window
2 × 2001, joint Ch1/Ch2 z-score
Ch1
Ch2
Standardized amplitude
Time from candidate (s)
ALS baseline estimation
λ = 10⁷, p = 0.01, per channel
ALS baseline (Ch1)
ALS baseline (Ch2)
Baseline removed, Ch2 scaled
Ch2 × σbg,Ch1 / σbg,Ch2
grey: excluded from σbg
(±0.15 s, edges)
dotted: central 0.2 s
Ch2 × σbg,Ch1/σbg,Ch2
(= 0.38)
Amplitude (a.u.)
Central 0.2 s × cosine taper
2 × 401, 0.025 s ramps
w
Ch2 · w
Ch1 · w
FFT magnitude
|rFFT| / Σw, Δf = 5 Hz
kept:
61 bins
discarded
> 300 Hz
|FFT| / Σw (a.u.)
Log magnitude, 0–300 Hz
61 bins, Δf = 5 Hz
log₁₀|Ch2|
log₁₀|Ch1|
log₁₀ magnitude
Cross-channel phase
cos(φCh1 − φCh2)
+1
in phase
0
−1
opposite
cos Δφ
Spectral-CNN input
61 × 3, z-scored per bin
z log|Ch1|
z log|Ch2|
cos Δφ
z-score / cos Δφ
Frequency (Hz)
→ spectral CNN (3 × Conv1D–BN–ReLU → Flatten → FC 64)